Interactive Learning Environments

ISSN: 1049-4820 (Print) 1744-5191 (Online) Journal homepage: www.tandfonline.com/journals/nile20

Enhancing postgraduate digital academic writing proficiency: the interplay of artificial intelligence tools and ChatGPT

Mohamed Oubibi, Katsiaryna Hryshayeva & Ronghuai Huang

To cite this article: Mohamed Oubibi, Katsiaryna Hryshayeva & Ronghuai Huang (2025) Enhancing postgraduate digital academic writing proficiency: the interplay of artificial intelligence tools and ChatGPT, Interactive Learning Environments, 33:6, 3940-3958, DOI: 10.1080/10494820.2025.2454445

To link to this article: https://doi.org/10.1080/10494820.2025.2454445

Published online: 24 Jan 2025.

Submit your article to this journal

Article views: 2045

View related articles

View Crossmark data

Citing articles: 29 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=nile20

INTERACTIVE LEARNING ENVIRONMENTS
2025, VOL. 33, NO. 6, 3940–3958
https://doi.org/10.1080/10494820.2025.2454445

Routledge
Taylor & Francis Group

Check for updates

Enhancing postgraduate digital academic writing proficiency: the interplay of artificial intelligence tools and ChatGPT

Mohamed Oubibi[a], Katsiaryna Hryshayeva[b] and Ronghuai Huang[a]

[a]Smart Learning Institute of Beijing Normal University, Faculty of Education, Beijing Normal University, Beijing, People's Republic of China; [b]School of Chinese Language and Literature, Beijing Foreign Studies University, Beijing, People's Republic of China

ABSTRACT

Artificial Intelligence (AI) has rapidly emerged as a pivotal force in a multitude of sectors. This research explores the growing convergence of AI tools, such as ChatGPT, in the field of academic writing, with a focus on 136 postgraduate students from China. These students were randomized into groups that utilized AI-integrated tools in both online and offline environments. The study involved designing hybrid instructional modules that blend online and offline elements. Furthermore, an extensive questionnaire was used to evaluate the advancements of postgraduate students in domains such as digital readiness, writing self-efficacy, student engagement, and academic emotional health. The results from this investigation highlight significant improvements in several facets of postgraduate involvement, including behavioral, emotional, and cognitive engagement, as well as writing self-efficacy. The study also probes the impact of digital readiness on academic involvement, revealing a positive association with student engagement, writing self-efficacy, and academic sentiments. By augmenting students' technical proficiency, digital literacy, and acclimation to digital tools and platforms, educational institutions can bolster academic engagement, amplify writing confidence, and foster favorable academic emotion.

ARTICLE HISTORY
Received 12 February 2024
Accepted 9 January 2025

KEYWORDS
Artificial intelligence; AI tools; generative AI; academic writing; ChatGPT; postgraduate students

1. Introduction

Academic writing plays a fundamental role in scholarly communication, research dissemination, and knowledge advancement. Effective academic writing requires a meticulous blend of clarity, coherence, and depth of analysis. However, many students and researchers encounter challenges in mastering this skill, ranging from organizing ideas coherently to adhering to rigorous academic standards and citation conventions (Selvaraj & Aziz, 2019; Storey, 2023). In recent years, the integration of Artificial Intelligence (AI) tools such as ChatGPT and other digital writing tools in the educational domain has paved the way for transformative solutions to address these challenges.

In today's interconnected academic environment, the ability to effectively communicate research in English is essential for global communication and collaboration. In addition, the exponential growth and advancement of AI technology have piqued people's interest in its application in the field of education (Phyo et al., 2023). Therefore, the scientific literature on the integration of AI into education (AIEd) has greatly expanded (Chen et al., 2022). AI has shown great potential in changing traditional teaching methods and providing personalized learning experiences and intelligent support to learners (Bhutoria, 2022).

CONTACT Mohamed Oubibi liby478@nenu.edu.cn
© 2025 Informa UK Limited, trading as Taylor & Francis Group

The purpose of this study is to explore the application of AI digital writing assistants in improving postgraduate academic participation, student participation, writing self-efficacy, and academic emotion. By investigating the impact of artificial intelligence technology based on AI tools and ChatGPT on these key factors, this study aims to contribute to AIEd's existing knowledge system and provide insights into how to use AI to support graduate students in academic writing. Therefore, the current study aims to address the following research questions:

The following issues raised concerns:

RQ1. Can AI-driven writing tools such as ChatGPT enhance postgraduate students' self-efficacy, engagement, and emotion?

RQ2. Can such an app serve as an effective tool for non-native graduate students to engage in academic writing in English?

2. Theoretical background

2.1. AI tools and academic writing

The use of AI technologies in academic writing has a significant positive impact on writing quality, academic integrity, and research output (Cotton et al., 2023). As AI technologies develop, continuing research and discussion are essential for maximizing their potential while ensuring that they are used ethically and responsibly in scholarly communication (Morley et al., 2020; Munoko et al., 2020; Prunkl et al., 2021). Personalized learning has been transformed by AI-driven systems, which now adapt instructional content to each student's unique needs, learning preferences, and speeds (Bhutoria, 2022). AI algorithms may create personalized learning pathways, suggest supplemental resources, and modify lesson plans in real time through data-driven analysis (Bhutoria, 2022). For diverse student populations, this makes learning more inclusive and efficient. An important paradigm shift in the way knowledge is transmitted and gained has been brought about by the use of AI tools in education (Alam, 2021). These tools, such as Grammarly, QuillBot, and ChatGPT, provide authors with immediate and accurate feedback to help them improve the clarity, coherence, and accuracy of their articles. The use of AI in grammar and style checking increases writing ability and improves the quality of manuscripts (Raheem et al., 2023).

AI tool's role extends beyond basic proofreading and plagiarism detection (Al Sawi & Alaa, 2024). Tools that assist in generating and organizing content, such as AI content generators and citation managers, aid students in structuring their ideas cohesively and referencing sources accurately (Hutson, 2024; Zhao, 2024). Additionally, AI-powered data analysis tools enable postgraduate students to analyze complex datasets efficiently, visualize findings, and derive meaningful insights for their research (Dahri et al., 2024). This capability is particularly beneficial in disciplines requiring rigorous empirical research, facilitating deeper exploration and interpretation of data. By leveraging AI tools effectively, postgraduate students can enhance their writing proficiency, elevate the quality of their research outputs, and ultimately achieve greater academic success in their respective fields (Borger et al., 2023; Malik et al., 2023).

2.2. Digital readiness and postgraduate student engagement

Digital readiness represents a multifaceted dimension of postgraduate students' readiness for not only academic achievement but also future professional accomplishments. It encompasses a diverse array of digital proficiencies, each with its unique perspective and attributes (Iglesias-Pradas et al., 2021; Kohnke & Moorhouse, 2021). Our investigation delved into the utilization of digital skills within the university setting and examined its association with academic involvement. Within the scope of our study, the term "digital" pertains to activities closely aligned with emerging information and communication technologies (Goodfellow & Lea, 2013). Student engagement, as

elucidated by Fredricks et al. in 2004, represents a meta-construct that encompasses three fundamental dimensions: behavioral, emotional, and cognitive engagement (Fredricks et al., 2004). The distinctive nature of engagement, which integrates these three dimensions, sets it apart from the extensive individual bodies of literature that only focus on each dimension separately. Behavioral engagement revolves around participation and encompasses involvement in academic, social, or extracurricular activities (Nguyen et al., 2018). It is a pivotal element for attaining positive academic outcomes and mitigating dropout rates (Banks & Dohy, 2019). Some scholars characterize it through positive behavior, which includes adherence to rules, conformity to classroom norms, and the absence of disruptive actions such as truancy or disciplinary issues (Calais & Green, 2022; Martins et al., 2022).

2.3. Students' self-efficacy writing

Self-efficacy refers to confidence in one's capacity to arrange and perform the required actions to accomplish specific objectives (Bandura, 1997). It holds significant sway over human behavior, impacting not only actions but also other emotional and motivational convictions. In the context of writing, self-efficacy relates to students' self-evaluations of their aptitude to proficiently execute writing assignments, encompassing self-perceived competencies such as grammar and mechanics (Katsarou, 2021; Mitchell et al., 2021).

Previous research has provided empirical evidence supporting the positive impacts of self-efficacy in writing (Ho et al., 2017). Recognizing the pivotal role of writing self-efficacy in influencing students' writing proficiency, it is crucial to devise a self-efficacy assessment tailored specifically to the realm of writing (An et al., 2021; Paul et al., 2021). Nevertheless, it is worth noting that the majority of self-efficacy assessments pertaining to writing have been formulated and implemented in the context of English as a first language (Teng et al., 2018).

2.4. Impact of emotions on student engagement and academic achievement

Academic emotion and student participation represent two closely intertwined constructs that have garnered significant attention within educational research (Bempechat & Shernoff, 2012). An in-depth understanding of the intricate relationship between these constructs is vital for advancing effective teaching and learning practices (Luan et al., 2023). Numerous studies have investigated the impact of academic emotions on students' cognitive and emotional processes (Wang et al., 2023). For instance, research conducted by Pekrun et al. revealed that experiencing enjoyment is positively correlated with students' intrinsic motivation, self-efficacy, and cognitive engagement. Conversely, feelings of anxiety are associated with reduced motivation, difficulties sustaining attention, and unfavorable learning outcomes (Pekrun et al., 2002).

Numerous investigations have revealed notable connections between students' pursuit of academic goals and the emotions they encounter in that pursuit (Clark et al., 2020; Oflaz, 2019). Within this framework, achievement emotions refer to the emotional states experienced by students before, during, or after engaging in academic tasks and experiencing their subsequent results (Camacho-Morles et al., 2021). These emotions encompass a spectrum from positive emotions such as enjoyment, pride, and hope to negative emotions such as boredom, anxiety, and anger. Moreover, these emotions can be characterized in terms of activation, with some, such as enjoyment, providing psychological stimulation, while others, such as boredom, have a deactivating impact (Cheng et al., 2023; Pekrun et al., 2023).

2.5. Transforming academic writing with AI tools and ChatGPT

The interplay of these AI tools contributes to enhanced academic writing by creating a comprehensive support system that addresses various aspects of the writing process (Dwivedi et al., 2023a; Lund et al.,

2023). For instance, while Grammarly focuses on grammar and style, QuillBot aids in paraphrasing and restructuring sentences, and ChatGPT offers content generation and idea expansion (Nurseha, 2023). When used together, these tools complement each other, allowing writers to not only produce grammatically correct and stylistically sound content but also to express their ideas in a coherent and well-structured manner (Elola & Oskoz, 2017). The collaborative use of these AI tools fosters a more efficient writing process, as each tool brings its strengths to the table, thereby reducing the time and effort required to produce high-quality academic work (Marzuki et al., 2023; Strobl et al., 2019). Moreover, the synergy between these AI tools can lead to improved learning outcomes. As students engage with these tools, they become more aware of their writing habits and areas where they need improvement (Abulibdeh et al., 2024; Rad et al., 2024). For example, Grammarly's feedback on sentence structure and tone may encourage a student to rethink their approach to academic writing, while QuillBot's paraphrasing suggestions can enhance their understanding of complex ideas (Calma et al., 2022; Fitria, 2021). ChatGPT's ability to provide contextual information and generate alternative viewpoints can stimulate critical thinking and deeper engagement with the subject matter (Qawqzeh, 2024). This interplay of AI tools not only enhances the quality of the writing but also contributes to the writer's overall development as a more skilled and reflective academic.

3. Materials and methods

3.1. Procedure

The inclusion criteria were as follows: participants who were more than 18 years old, who were enrolled as full-time graduate students at a Chinese university, and who possessed acceptable digital literacy. We employed a multimodal targeting strategy to reach participants through popular Chinese social media platforms such as WeChat groups, QQ, and email. Participants took part in an online survey. The survey process began in September 2022 and the second part was completed in January 2023. We collected a specific sample because most students were busy attending classes and coping with the COVID-19 pandemic. Before responding, every participant received explicit assurance of anonymity, and the researchers pledged unwavering confidentiality throughout the process of analysis and reporting. Additionally, participants were informed that the gathered information would be exclusively employed for designated academic research purposes. Only those who willingly consented to participate in the study were invited to proceed with the assessment agreement. To ensure adherence to the inclusion criteria, participants had the option to provide their student number, an indicator of their academic proficiency level. Notably, all the surveys were completed by participants within a time frame of approximately 10 to 15 minutes, and no form of incentive was issued for their participation. The research protocol obtained formal approval from the ethical committee at X University's School of Education and rigorously adhered to the ethical tenets outlined in the Helsinki Declaration.

3.2. Participants

In this study, we collected data from full-time graduate students. The final sample consisted of 136 graduate students from different universities in China, with 79 (58.08%) pursuing a master's degree and 57 (41.92%) enrolled as doctoral students. Their ages ranged from 20 to 45 years old, and they had varying levels of teaching experience. More than 80% of the graduate students had less than five years of experience, while only 6% had over ten years of experience.

3.3. Instruments

3.3.1. AI-powered digital tools

OpenAI created the language model known as ChatGPT. ChatGPT is built to comprehend and produce text in a conversational fashion, it may be used for a variety of tasks, such as answering

inquiries, providing explanations, creating content, and making recommendations. It has the capacity to converse with users and produce responses that are pertinent to the current situation.

Grammarly, an AI tool introduced by Taguma and team in 2018, constitutes a versatile tool in the domain of writing. It offers automated writing evaluation (AWE), automated essay scoring (AES), and automated writing feedback (AWCF) applications within a digital framework.

QuillBot is a popular supplementary paraphrase tool. Dwivedi et al. (2023) claimed that it is a market-available, reasonably priced paraphrase solution that uses cutting-edge AI technology to successfully rewrite any provided content (Dwivedi et al., 2023).

3.3.2. Digital readiness

The Digital Readiness Assessment and Evaluation (DRAE), devised by Hong and Kim in 2018, is a self-report tool comprising 17 items (Hong & Kim, 2018). It is tailored to appraise digital readiness and students' self-perceived proficiency on five distinct subscales linked to academic engagement. In our investigation, students conveyed their levels of agreement with five statements (1 = strongly disagree, 5 = strongly agree). The scores obtained from these responses serve as indicators of both digital readiness and students' perceived digital competencies within the realm of academic engagement. Remarkably, the scale exhibited robust reliability, with a Cronbach's alpha coefficient of $\alpha = 0.91$ in this particular study.

3.3.3. Student engagement

The Student Engagement Scale, originally introduced by Fredricks and team in 2005, is a self-reported instrument comprising 19 items. It evaluates student engagement across three distinct sub-dimensions. For our study, we utilized five items centered on behavior (e.g. "I am committed to adhering to the guidelines of this online course"). Furthermore, emotional engagement was assessed using six items (e.g. "I take pleasure in actively participating in this online course"), while cognitive engagement was assessed using eight items (e.g. "I am determined to grasp the content of this online course, even in moments of difficulty"). Participants expressed their levels of agreement through a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). It is worth noting that the internal consistency, as gauged by Cronbach's alpha, demonstrated commendable reliability across the three subdimensions: 0.81 for behavior, 0.77 for emotion, and 0.79 for cognitive engagement.

3.3.4. Self-Efficacy writing scale

The Self-Efficacy Writing Scale (SEWS), which was devised by Bruning and colleagues in 2013, comprises 16 self-reported items employed for gauging one's confidence in the domain of writing. This scale encompasses five items focused on idea generation, another five concerning conventions, and an additional six addressing self-regulation. Students were prompted to articulate their levels of concurrence with five distinct statements, employing a scale ranging from 1 (strongly disagree) to 5 (strongly agree). Notably, the internal consistency within the present study demonstrated commendable reliability ($\alpha = 0.81$).

3.3.5. Achievement emotion questionnaire

The Achievement Emotion Questionnaire (AEQ), developed by Pekrun and colleagues in 2012, serves as the tool of choice for gauging both positive and negative emotional states (Pekrun & Linnenbrink-Garcia, 2012). This instrument delves into emotions within three distinct contexts. Within the scope of this investigation, students reflected upon their emotional experiences while they were learning English, which was further divided into two subdimensions. To explore positive emotions, three items centered on enjoyment (e.g. "I find pleasure in acquiring new knowledge about writing research in English") were employed. Four items focused on anxiety (e.g. "I experience nervousness and tension when composing research in English") were used to probe negative emotions. The respondents indicated their agreement on a five-point Likert scale ranging from 1 (strongly disagree)

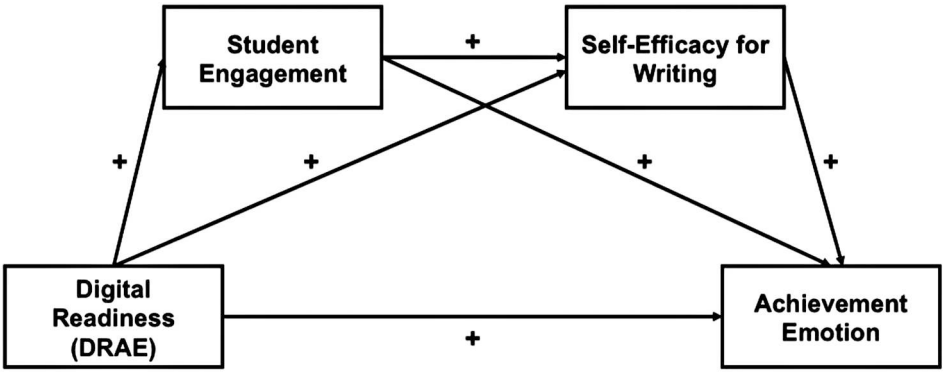


Figure 1. Multiple regression model for the effect of digital readiness, student engagement, and self-efficacy for writing on achievement emotion.

to 5 (strongly agree). It is noteworthy that the internal consistency observed in this study demonstrated a commendable level of reliability ($\alpha = 0.83$).

3.4. Research model

Based on our research plan and the model depicted in Figure 1, the following model was used in this study. This model describes the relationships between DRAE, Student Engagement (SE), SEWS, and AEQ. More details about the research model are shown in Figure 1.

This study used our research model to verify whether digital readiness affects students' self-efficacy for writing, achievement emotion, student engagement on self-efficacy for writing and achievement emotion, and self-efficacy writing on achievement emotion.

3.5. Content of the experiment

The experiment was conducted online and offline over twelve weeks (Table 1), with three- hour classes on Sundays. All participants completed a survey before starting the experiment, and informed consent was obtained from the subject. We informed the subject that all the information gathered in this study would be used only for a specific academic study.

4. Results

4.1. Descriptive statistics

In this study, 136 participants took part in the study in both offline (62, 45.58%) and online (74, 54.42%), aged between 20 and 45 years old, from different countries, and had different levels of education (master's degree, 79, 58.08% and doctorate degree, 57, 41.92%) participated in the study. More details are provided in Table 2.

4.2. Factor loading and validation

4.2.1. DRAE factor loading and validation

During analysis of the collected data, factor loading and validation were checked for digital readiness (Table 3), and Factor 1 encompassed four items each with factor loadings between 0.770 and 0.860. These items exclusively pertain to students' utilization of digital tools in their academic endeavors. Consequently, Factor 1 is denoted as Digital Tool Application (DTA).

Table 1. Content of the classes during the experiment.

Session	Content of Session	Objectives	Homework
Week One	Presentation: Writing tools (Generative AI, Grammarly, ChatGPT, Quillbot …). EndNote, Mendeley, Ps, AMOS.	Introducing students to a suite of writing tools Enhances their writing accuracy and creativity.	Old article or prepare article.
Week Two	Find articles, report, thesis. From where to read.	Students will be well-equipped to navigate the academic landscape, access relevant materials, and present their own research and findings in a structured and compelling manner.	Write an APA template.
Week Three	Prepare Manuscript – tools Structuring article.	Professional understanding of data tools necessary for different manuscript.	Prepare an article.
Week Four	Article structure – Data collection – Data Analysis (SPSS-JASP-AMOS software's …).	Type of thesis and articles. Adequate software and methods.	Write an Abstract.
Week Five	Use of ChatGPT, Grammarly, QuillBot …	Effective use of AI powered digital writing assistance.	Prepare graduated thesis.
Week Six	Thesis to Article	Prepare a draft from previous bachelor or master thesis.	Bachelor/Master Thesis to art.
Week Seven	Cover letter – Email to editor.	Prepare students to write professional letter and emails.	Write a cover letter and Email.
Week Eight	Check manuscript plagiarism (Turnitin …)	Students will be able to check manuscript plagiarism and paraphrase documents and reduce plagiarism.	Find plagiarism in report.
Week Nine	Report Data, References	Students will be able to analyze data with SPSS, AMOS, JASP and download references.	Write a standard report.
Week Ten	Publisher, journal, conference Find convenient journal-based title – Abstract.	Students will be able to select convenient journal and understand the abstracting of journals.	Find a list of SSCI journals in the field of educational technology.
Week Eleven	Proofreading, Mistake to avoid during checking and submission	Professional understanding of online submission system.	Revise an article.
Week Twelve	Revise a submitted manuscript	Understanding the difference between writing and reviewing manuscript.	Submit manuscript.

Factor 2 comprises four items, with factor loadings ranging from 0.722 to 0.796. Each of these items pertains to students' proficiency in media literacy within academic contexts. Factor 2 is identified as Information Sharing behavior (ISB).

4.2.2. Student engagement factor loading and validation

For the student engagement factor leading (Table 4), Factor 1 encompasses five items, with factor loadings ranging from 0.642 to 0.765. Each of these items exclusively pertains to students' behaviors within the realm of academic work. Consequently, Factor 1 designated as Behavior (BE).

Table 2. Descriptive statistics (n = 136)

N	Attribute	Type	Frequency	Proportion
1	Gender	Male	59	43.38%
		Female	77	56.61%
2	Age	20–25	83	61.03%
		26–35	38	27.94%
		36–45	15	11.03%
3	Education	Master	79	58.08%
		Ph.D.	57	41.92%
4	Participant	Offline	62	45.58%
		Online	74	54.42%
5	Experience	0–2	83	61.02%
		2–5	31	22.79%
		5–10	13	9.55%
		+10	9	6.61%

Table 3. Digital readiness factor loadings.

Factor	Indicator	Estimate	Std.Error	95% Confidence Interval		
				p	Lower	Upper
Digital tool Application	DTA-1	0.784	0.009	< .001	0.767	0.702
	DTA-2	0.770	0.008	< .001	0.754	0.786
	DTA-3	0.860	0.008	< .001	0.845	0.876
	DTA-4	0.835	0.008	< .001	0.819	0.852
Information Behavior	ISB-1	0.796	0.016	< .001	0.765	0.828
	ISB-2	0.722	0.012	< .001	0.699	0.745
	ISB-3	0.777	0.014	< .001	0.750	0.804
ISS	ISS-1	0.781	0.014	< .001	0.754	0.808
	ISS-2	0.802	0.013	< .001	0.776	0.829
	ISS-3	0.644	0.017	< .001	0.612	0.676
Digital Media Awareness	DMA-1	0.765	0.018	< .001	0.730	0.800
	DMA-2	0.777	0.019	< .001	0.741	0.814
	DMA-3	0.883	0.022	< .001	0.840	0.927
Digital Application Usage	DAU-1	0.685	0.013	< .001	0.660	0.709
	DAU-2	0.724	0.013	< .001	0.698	0.750
	DAU-3	0.788	0.013	< .001	0.763	0.714
	DAU-4	0.765	0.013	< .001	0.739	0.791

Factor 2 comprises six items, with factor loadings ranging from 0.666 to 0.708. All six of these items specifically address students' emotional experiences in academic endeavors. Consequently, Factor 2 is identified as Emotion (EMO).

Factor 3 encompasses eight items, demonstrating factor loadings between 0.670 and 0.790. All six of these items are centered on students' capacity for self-regulation in academic writing. Hence, Factor 3 is labeled Cognition (COG).

These factors, BE, EMO, and COG, were meticulously delineated based on their respective item content and factor loadings. They provide valuable insights into distinct aspects of students' engagement in academic work.

Table 4. Student engagement factor loadings.

Factor	Indicator	Estimate	Std.Error	95% Confidence Interval		
				p	Lower	Upper
Behavioral	BE-1	0.669	0.056	< .001	0.658	0.679
	BE-2	0.730	0.060	< .001	0.713	0.746
	BE-3	0.765	0.057	< .001	0.752	0.777
	BE-4	0.642	0.066	< .001	0.512	0.771
	BE-5	0.753	0.058	< .001	0.638	0.867
Emotional	EMO-1	0.708	0.062	< .001	0.686	0.830
	EMO-2	0.674	0.055	< .001	0.566	0.781
	EMO-3	0.741	0.059	< .001	0.626	0.856
	EMO-4	0.754	0.083	< .001	0.591	0.918
	EMO-5	0.666	0.098	< .001	0.673	0.759
	EMO-6	0.691	0.059	< .001	0.676	0.707
Cognitive	COG-1	0.784	0.064	< .001	0.658	0.909
	COG-2	0.790	0.077	< .001	0.740	0.840
	COG-3	0.708	0.065	< .001	0.736	0.780
	COG-4	0.673	0.070	< .001	0.536	0.710
	COG-5	0.759	0.065	< .001	0.687	0.832
	COG-6	0.670	0.062	< .001	0.649	0.690
	COG-7	0.726	0.096	< .001	0.714	0.761
	COG-8	0.021	0.102	< .001	−0.220	0.179

Table 5. Self-efficacy writing factor loadings.

				95% Confidence Interval		
Factor	Indicator	Estimate	Std.Error	*p*	Lower	Upper
Ideation	IDE-1	0.884	0.009	< .001	0.867	0.901
	IDE-2	0.871	0.008	< .001	0.855	0.886
	IDE-3	0.760	0.008	< .001	0.744	0.775
	IDE-4	0.835	0.008	< .001	0.819	0.852
	IDE-5	0.862	0.016	< .001	0.831	0.893
Convention	C-1	0.768	0.013	< .001	0.743	0.794
	C-2	0.853	0.015	< .001	0.724	0.882
	C-3	0.833	0.018	< .001	0.799	0.867
	C-4	0.835	0.018	< .001	0.801	0.869
	C-5	0.816	0.018	< .001	0.781	0.850
Self-regulation	SR-1	0.791	0.015	< .001	0.762	0.821
	SR-2	0.766	0.015	< .001	0.735	0.796
	SR-3	0.987	0.018	< .001	0.951	1.023
	SR-4	0.837	0.016	< .001	0.805	0.869
	SR-5	0.930	0.018	< .001	0.894	0.965
	SR-6	0.708	0.013	< .001	0.682	0.733

4.2.3. SEW factor loading and validation

For the self-efficacy writing factor leading (Table 5), Factor 1 comprised five items, exhibiting factor loadings between 0.760 and 0.884. All of these items exclusively pertain to students' generation of ideas within the context of academic work. Consequently, Factor 1 is aptly labeled Idea (IDE).

Factor 2 encompasses five items, demonstrating factor loadings spanning from 0.768 to 0.853. Each of these five items specifically addresses students' adherence to conventions in their academic endeavors. Consequently, Factor 2 is citified as Convention (C).

Factor 3 consists of six items, displaying factor loadings between 0.766 and 0.987. Each of these six items is centered on students' capacity for self-regulation in their academic writing. Hence, Factor 3 is aptly labeled self-regulation (SR).

These factors, IDE, C, and SR, have been meticulously delineated based on their respective item content and factor loadings. They serve as valuable indicators of distinct facets of students' engagement in academic work.

4.2.4. Achievement emotion factor loading and validation

For achievement emotion factor leading (Table 6), Factor 1 comprises eight items that had factor loadings between 0.533 and 0.856. Factor 1 (CRE) is aptly labeled as class-related emotion.

Factor 2 also comprises eight items labeled (LRE) for learning-related emotion and had factor loadings between 0.548 and 0.676, and Factor 3 labeled (TE) for test emotion; this factor also comprises eight factors, leading to loadings between 0.611 and 0.774.

These factors, CRE, LRE, and TE, have been meticulously delineated based on their respective item content and factor loadings. They serve as valuable indicators of distinct facets of postgraduate students' achievement emotions.

4.3. Correlation

Pearson correlation analysis revealed favorable associations among our main variables DRAE, SE, SEW, and AE. Additionally, age, educational level, and teaching experience showed favorable correlations with other factors; for more details, refer to Table 7.

The Table 7 shows that there are positive correlations among the main variables (digital readiness, student engagement, self-efficacy for writing, and achievement emotion) with educational level, experience, and age.

Table 6. Achievement emotion factor loadings.

				95% Confidence Interval		
Factor	Indicator	Estimate	Std.Error	*p*	Lower	Upper
Class related-emotion	CRE-1	0.853	0.015	< .001	0.724	0.882
	CRE-2	0.682	0.008	< .001	0.666	0.697
	CRE-3	0.671	0.008	< .001	0.655	0.686
	CRE-4	0.856	0.070	< .001	0.718	0.994
	CRE-5	0.762	0.071	< .001	0.623	0.900
	CRE-6	0.517	0.095	< .001	0.330	0.704
	CRE-7	0.533	0.090	< .001	0.356	0.710
	CRE-8	0.835	0.008	< .001	0.819	0.852
Learning related-emotion	LRE-1	0.645	0.064	< .001	0.520	0.770
	LRE-2	0.624	0.060	< .001	0.506	0.741
	LRE-3	0.548	0.058	< .001	0.434	0.662
	LRE-4	0.604	0.056	< .001	0.494	0.714
	LRE-5	0.676	0.054	< .001	0.570	0.781
	LRE-6	0.634	0.055	< .001	0.526	0.742
	LRE-7	0.657	0.056	< .001	0.548	0.767
	LRE-8	0.559	0.055	< .001	0.781	0.850
Test emotion	TE-1	0.649	0.078	< .001	0.497	0.801
	TE-2	0.720	0.069	< .001	0.585	0.855
	TE-3	0.686	0.056	< .001	0.575	0.797
	TE-4	0.735	0.068	< .001	0.602	0.868
	TE-5	0.774	0.062	< .001	0.651	0.896
	TE-6	0.735	0.062	< .001	0.614	0.856
	TE-7	0.708	0.061	< .001	0.588	0.828
	TE-8	0.611	0.100	< .001	0.514	0.707

Table 7. Pearson's correlations between the main variables.

Variables		Learning degree	Experience	Age	DRAE	SE	SEW	AE'
1.Learning degree	Pearson's r	–						
	p-value	–						
2.Experience	Pearson's r	0.725	–					
	p-value	<.001	–					
3.Age	Pearson's r	0.750	0.963	–				
	p-value	< .001	<.001	–				
4.DRAE	Pearson's r	0.072	0.053	0.076	–			
	p-value	0.406	0.542	0.377	–			
5.SE	Pearson's r	0.026	0.111	0.134	0.702	–		
	p-value	0.760	0.200	0.120	< .001	–		
6.SEW	Pearson's r	0.034	0.048	0.069	0.013	0.143	–	
	p-value	0.691	0.577	0.422	0.885	0.096	–	
7.AE	Pearson's r	−0.025	−0.002	−0.011	0.268	0.325	−0.087	–
	p-value	0.777	0.977	0.898	0.002	< .001	0.316	–

4.4. Confirmatory factor analysis

Confirmatory Factor Analysis (CFA) was performed to evaluate the discriminant validity of our measurement model regarding academic emotion. This process involved several CFA indices, including the Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), Goodness-of-Fit Index (GFI), and Expected Cross-Validation Index (ECVI), as outlined in Table 8. This comprehensive examination confirmed the robustness and accuracy of our measurement model in relation to academic emotion.

4.6. Multiple mediation models

To test the paths among the main variables, process macro-Model 6 was used to test multiple mediation models. In summary, all pathway coefficients were found to be significant. More details

Table 8. CFA results.

Index	DRAE	SE	SEW	AE
Comparative Fit Index (CFI)	0.943	0.930	0.942	0.903
Tucker-Lewis Index (TLI)	0.929	0.919	0.913	0.892
Bentler-Bonett Non-normed Fit Index (NNFI)	0.929	0.919	0.913	0.892
Root mean square error of approximation (RMSEA)	0.079	0.095	0.133	0.089
RMSEA *p*-value	0.000	0.000	0.000	0.000
Standardized root mean square residual (SRMR)	0.066	0.032	0.101	0.054
Goodness of fit index (GFI)	0.993	0.985	0.984	0.973
McDonald fit index (MFI)	0.712	0.808	0.409	0.375
Expected cross validation index (ECVI)	0.753	1.439	1.853	2.087

about the results of the multiple mediation models concerning digital readiness, student engagement, self-efficacy writing and academic emotion can be found in Table 9, which displays multiple mediation models.

The pathways "Digital Readiness → Student Engagement → Academic Emotions" (indirect effect = 0.123, 95% CI = [0.025, 0.271]) and "Digital Readiness → Self-Efficacy Writing → Academic Emotions" (indirect effect = 0.131, 95% CI = [0.009, 0.253]) were statistically significant. This indicates that student engagement and self-efficacy in writing concurrently play a mediating role in the relationship between digital readiness and academic emotions. Additionally, the sequential pathway "Digital Readiness → Student Engagement → Self-Efficacy Writing → Academic Emotions" also showed significance (indirect effect = 0.253, 95% CI = [0.072, 0.435]). This suggests that heightened digital readiness is positively linked with increased student engagement (b = 0.363, $p < 0.001$), academic emotions (b = 0.305, $p < 0.001$), and self-efficacy in writing (b = 0.271, $p < 0.001$). Furthermore, the direct pathway "Student Engagement → Academic Emotions" was found to be significant (b = 0.338, $p < 0.001$, 95% CI = [0.009, 0.551]), as was the direct pathway from Student Engagement to Self-Efficacy Writing (b = 0.022, $p < 0.001$, 95% CI = [0.008, 0.051]). Therefore, both concurrently and sequentially, student engagement and self-efficacy writing mediate the relationship between digital readiness and academic emotions. Overall, this multiple mediation model explains a substantial portion of the variance in graduate students' academic emotional changes. Figure 2 displays the linear regression analysis of our model on the main variables.

4.7. Pre-test and post-test

This study employed a pre-test and post-test design to assess the baseline characteristics of the graduate students. It served as one of the primary techniques for researchers to compare participant groups and to measure the extent of change brought about by the treatment or intervention at the

Table 9. Direct, indirect, and total effects of multiple mediation (n = 136).

Effect	B	SE	95% Boot LLCI	95% Boot ULCI
		Direct effects		
DRAE → SE	0.363	0.048	0.270	0.456
DRAE → SEW	0.305	0.051	0.206	0.404
DRAE → AE	0.271	0.143	0.009	0.551
SE → AE	0.338	0.203	0.060	0.737
SEW → AE	0.429	0.191	0.054	0.804
SE → SEW	0.022	0.015	0.051	0.051
		Indirect effects		
DRAE → SE → AE	0.123	0.076	0.025	0.271
DRAE → SEW → AE	0.131	0.062	0.009	0.253
DRAE →SE → SEW → AE	0.253	0.093	0.072	0.435

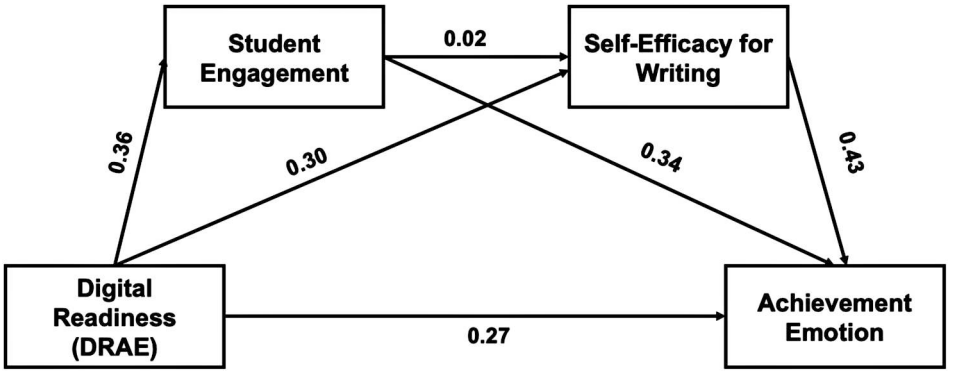


Figure 2. Linear regression analysis of the main variables.

end of the experiment (after one semester of study) to gauge the elements mentioned in the learning objectives (Table 10).

Table 10 shows that all key variables (digital readiness, student engagement, self-efficacy writing, and academic emotions) improved between the pre-test and post-test. For example, the average score for digital readiness in the pre-test (N = 136, M = 1.922*) increased to (N = 136, M = 3.822) in the post-test. In the pre-test (N = 136, M = 3.019**) and post-test (N = 136, M = 3.956**) experiments, student engagement scores were enhanced. Similarly, there was an increase in self-efficacy in writing, as evident from the pre-test (N = 136, M = 1.063**) to the post-test (N = 136, M = 4.212*). This trend was also observed for graduate students' academic emotions, with pre-test scores (N = 136, M = 1.634**) increasing to (N = 136, M = 4.271**) in the post-test.

4.8. Paired sample T test

Pretests were administered to graduate students at the start of the experiment to establish their initial understanding of the measures outlined in the learning objectives. Post-tests, on the other hand, were conducted after the completion of the course to determine the skills acquired and developed by graduate students in academic writing and publishing. In this section, of the experiment, paired sample t tests were used to determine whether there was sufficient evidence to conclude that the average difference between paired observations for a given outcome was significantly different from zero. Table 11 provides detailed comparisons of means from the same individuals, items, or related units.

The paired sample t tests revealed substantial disparities between the pre-test and post-test scores for each variable. Specifically, regarding digital readiness, there was a considerable difference between the pre-test and post-test scores (t135 = 39.34, $p < 0.001$). The mean difference between post-test and pre-test scores for digital readiness was 1.90** (95% CI: 1.79**, 1.99**).

Table 10. Paired sample statistics.

V		M	N	SD	SE
DRAE	Pretest	1.922*	136	0.320	0.01
	Posttest	3.822**	136	0.480	0.01
SE	Pretest	3.019**	136	0.255	0.02
	Posttest	3.956**	136	0.479	0.01
SEW	Pretest	1.063**	136	0.275	0.02
	Posttest	4.212*	136	0.312	0.01
AE	Pretest	1.634**	136	0.226	0.01
	Posttest	4.271**	136	0.405	0.01

Table 11. Postgraduate student paired samples with t tests.

		Paired Differences					t	df	Sig.
					95% CID				
MW	Test	M	SD	SE	Lower	Upper			
DRAE	Pretest Posttest	1.90**	0.54**	0.03**	1.79**	1.99**	39.34	135	0.000
SE	Pretest Posttest	0.93**	0.54**	0.02**	0.83**	1.04**	37.46	135	0.000
SEW	Pretest Posttest	3·14**	0.40**	0.03**	3.05**	3.24**	48.53	135	0.000
AE	Pretest Posttest	2.63**	0.46**	0.03**	2.52**	2.73**	47.95	135	0.000

5. Discussion and limitations

The findings of this study highlight the multifaceted role of digital readiness in shaping academic emotions among graduate students, mediated through student engagement and self-efficacy in writing. These results align with existing literature while offering novel insights into the intricate relationships between these constructs, thus contributing to the broader understanding of digital literacy's impact on higher education. Among the various factors influencing student academic achievement, the development of students' emotional intelligence has received particular attention because it contributes to their ability to reflect and regulate their emotional states, further enhancing their intellectual development. Our findings confirm that digital readiness positively impacts academic emotions through enhanced student engagement and writing self-efficacy. This is consistent with Goode (2010), who emphasized the necessity of digital readiness for academic success in modern education landscapes. The academic landscape has changed, with students being expected to use online resources for research, complete multimedia assignments, navigate learning management systems, and submit assignments through email, among other digital tools and applications. The first set of results indicates that student engagement and self-efficacy for writing serve as mediating factors between digital readiness and academic emotions. This suggests that when graduate students are digitally prepared, their engagement in academic activities and their beliefs about their writing abilities are positively influenced, thereby enhancing their academic emotions. The moderate size of the indirect effect (0.123, confidence interval [0.025, 0.271]) suggested a moderate association between digital readiness, student engagement, and academic emotions. This supports the existing body of literature highlighting the a moderate association between digital readiness, students engagement and academic emotions (Händel et al., 2020).

Similarly, the second set of results demonstrates the indirect impact of digital readiness on academic emotions through self-efficacy writing. In other words, digitally prepared graduate students are more likely to have higher self-efficacy in writing, resulting in more positive academic emotions. The moderate size of the indirect effect (0.131, confidence interval [0.009, 0.253]) implies a moderate association between digital readiness, self-efficacy for writing, and academic emotions. This finding aligns with previous research in the field of academic self-efficacy and emotions for writing (Cassaretto et al., 2024; Tecedor, 2024). Furthermore, the third set of results reveals the significance of a sequential mediation model involving digital readiness, student engagement, self-efficacy for writing, and academic emotions. This indicates that digital readiness not only directly influences student engagement, academic emotions, and self-efficacy for writing but also sequentially influences these variables. The moderate size of the indirect effect (0.253, confidence interval [0.072, 0.435]) suggests a relatively strong association among these variables. This findings align with the observations made by Getenet et al. (2024) on preservice teachers' digital readiness influences efficacy, attitude and engagement and emotional engagement (Getenet et al., 2024). Moreover, the results revealed a significant direct path from student engagement to academic emotions, indicating that higher levels of student engagement have a positive impact on academic emotions. The

positive coefficient of 0.338 (confidence interval [0.009, 0.551]) indicates a significant influence of student engagement on academic emotions.

It is important to acknowledge that these findings are based on a specific sample and context of the study. Replication studies and further research in different educational settings are necessary to generalize these results and ascertain the robustness of the identified relationships. Nonetheless, these findings provide valuable insights into the interplay between graduate students' digital readiness, student engagement, self-efficacy for writing, and academic emotions, offering a foundation for guiding educational practices and policy-making (Greenhow et al., 2022). The study findings underscore the value of employing the DRAE scale as a universal tool for gaining deeper insights into the collective influence of digital literacy on student engagement across diverse academic learning endeavors. For instance, educators looking to seamlessly integrate technology into their classrooms can leverage the DRAE scale to glean valuable insights into how students interact with digital technologies during learning activities. With this knowledge, instructors can refine their teaching strategies and tailor resources to yield concrete educational outcomes, thus better addressing the unique learning needs of their students. Furthermore, our results shed light on the efficacy of AI as an intervention in bolstering the acceptance factors outlined in our second hypothesis. The data demonstrate that AI serves as a potent catalyst for enhancing second-grade students' self-efficacy and their emotional engagement in academic writing. These findings align with prior research on the effectiveness of AI in second language (L2) learning among students (Cotos, 2011; Parra & Calero, 2019). In the realm of English writing, students endowed with higher levels of self-efficacy are inclined to invest more effort and persist, ultimately yielding superior writing outcomes (Usher & Pajares, 2008).

There were significant average differences in student engagement scores between pre-test and post-test. This indicates that the use of an AI writing assistant has a positive impact on graduate students' motivation, interest, and active participation in academic tasks and the following finding align with the study by Song & Song, 2023 on the positive impact of AI writing tools enhance academic writing skills (Song & Song, 2023). This finding aligns also with previous research, highlighting the potential of technology-enhanced interventions in promoting student engagement, which is a key factor for academic success (Bond et al., 2020; Lee et al., 2022). For self-efficacy in writing, the significant average differences between the pre-test and post-test self-efficacy writing scores indicate that AI writing assistants have a significant influence on graduate students' confidence and beliefs in their writing abilities. The personalized feedback, guidance, and assistance provided by the tool may contribute to the development of self-efficacy, thereby enhancing writing performance. This finding underscores the potential of artificial intelligence technology in fostering self-efficacy and empowering students to become more effective writers.

Several studies demonstrated how Grammarly as AI-powered English writing (Fitria, 2021), Quillbot (Amanda et al., 2023), ChatGPT (Lingard, 2023) to support and assist students in proofreading, paraphrasing, summarizing and improving their skills in academic writing. This study contributes meaningfully to the educational community by providing actionable insights into the mechanisms linking digital literacy, engagement, and academic emotions. The validation of the DRAE scale as a tool for assessing digital readiness offers educators and policymakers a practical means of gauging and enhancing student engagement across diverse learning contexts. Furthermore, the demonstrated efficacy of AI tools suggests their potential as scalable interventions to support students with varying levels of digital and emotional readiness. Beyond immediate pedagogical applications, the study lays the groundwork for future research to explore additional variables, such as the quality of digital resources, prior experience with technology, and individual differences in learning styles. Such inquiries can enrich the understanding of digital readiness and its broader implications, ultimately guiding educators in creating more inclusive and effective learning environments. These findings not only reaffirm the critical role of digital readiness in education but also suggest innovative pathways for integrating technology to improve academic outcomes.

6. Conclusion

Research on AI-driven digital writing assistants has revealed the impact of digital readiness on various critical aspects of graduates' academic experiences. The study results offer academic and scientific insights into the relationships between digital readiness, academic engagement, student participation, writing self-efficacy, and academic emotions in the graduate context. By elucidating the mediating role of student engagement and self-efficacy in connecting digital readiness with academic emotions during writing, this research contributes to the theoretical understanding of digital readiness. The findings enhance our understanding of the sequential mediation model, emphasizing the cumulative impact of digital readiness, student engagement, writing self-efficacy, and academic emotions on graduate students' emotional experiences. From a practical perspective, this study holds significance for educational institutions. This highlights the need for specialized digital readiness programs and interventions tailored to graduate students. By enhancing students' technical skills, digital literacy, and familiarity with digital tools and platforms, institutions can foster positive academic engagement, boost writing self-efficacy, and cultivate positive academic emotions.

This study contributes to the theoretical understanding of the relationships among digital readiness, student engagement, and academic emotions. This study highlights the mediating roles of student engagement and self-efficacy in writing in connecting digital readiness with academic emotions. This study provides insights into the potential mechanisms through which digital readiness influences students' emotional experiences in academic writing. This study introduces a sequential mediation model that incorporates digital readiness, student engagement, self-efficacy for writing, and academic emotions. This model helps elucidate the sequential nature of these variables and their cumulative effects on graduate students' academic emotions. The theoretical framework and model can be further refined and expanded by incorporating these sequential mediation processes into existing theories of student engagement, self-efficacy, and emotional experiences in educational environments.

While exploring the interplay of AI tools in enhancing postgraduate digital academic writing proficiency, several limitations should be considered. The generalizability of findings may be constrained by the specific focus on certain AI tools or academic disciplines. Different fields may have distinct writing norms and requirements that AI tools might not universally address. Moreover, the study's applicability could vary depending on the technological infrastructure and access to AI resources available in different educational settings. Disparities in internet connectivity, access to advanced AI technologies, and varying levels of technical proficiency among students could impact the study's broader relevance across diverse academic environments. Additionally, while AI tools can enhance writing mechanics, they may not necessarily promote critical thinking or creativity, which are essential skills in academic writing. Therefore, understanding the nuanced impacts of AI on skill development and educational outcomes among postgraduate students requires comprehensive exploration beyond technical proficiency alone.

Contributions

MO Conceptualization, formal analysis, methodology, and writing the original manuscript, analyzed and interpreted the data.

KH Conceptualization, formal analysis, methodology, and writing the original manuscript.

RH Conceptualization, formal analysis, methodology, and writing the original manuscript.

Declarations

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Mohamed Oubibi is a Lecturer and research fellow at Smart Learning Institute of Beijing Normal University, China, and Visiting Professor at Rattana Bundit University, Thailand. His research interests include Smart education, AI, AR, VR, ICT in education, big data, teacher competencies, and educational technology.

Katsiaryna Hryshayeva is a Ph.D. Candidate at Beijing Foreign Studies University majoring in Teaching Chinese as a Foreign Language, her research interest Include, Smart Education, AI in education, TCFL, and Teacher competenies.

Ronghuai Huang is a Professor in Beijing Normal University. He has being engaged in the research on smart learning environment, artificial intelligence in education, and educational technology. He serves as Co-Dean of Smart learning Institute, UNESCO Chair on AIED, and Director of China National Engineering Research Center for Cyber-learning and Intelligent Technology, and Editor-in-Chief of Springer's Journal of Smart Learning Environment and Journal of Computers in Education.

Data availability statement

Data will be made available on request.

References

Abulibdeh, A., Zaidan, E., & Abulibdeh, R. (2024). Navigating the confluence of artificial intelligence and education for sustainable development in the era of industry 4.0: Challenges, opportunities, and ethical dimensions. *Journal of Cleaner Production*, *437*, 140527. https://doi.org/10.1016/j.jclepro.2023.140527.

Alam, A.. (2021). *Possibilities and apprehensions in the landscape of artificial intelligence in education* [Conference session]. 2021 International Conference on Computational Intelligence and Computing Applications (ICCICA), Nagpur, India (pp. 1–8). https://doi.org/10.1109/ICCICA52458.2021.9697272.

Al Sawi, I., & Alaa, A. (2024). Navigating the impact: A study of editors' and proofreaders' perceptions of AI tools in editing and proofreading. *Discover Artificial Intelligence*, *4*(1), 23. https://doi.org/10.1007/s44163-024-00116-5

Amanda, A., Sukma, E. M., Lubis, N., & Dewi, U. (2023). Quillbot as an AI-powered English writing assistant: An alternative for students to write English. *Jurnal Pendidikan dan Sastra Inggris*, *3*(2), 188–199. https://doi.org/10.55606/jupensi.v3i2.2026

An, Z., Wang, C., Li, S., Gan, Z., & Li, H. (2021). Technology-assisted self-regulated English language learning: Associations with English language self-efficacy, English enjoyment, and learning outcomes. *Frontiers in Psychology*, *11*, 558466. https://doi.org/10.3389/fpsyg.2020.558466

Bandura, A. (1997). *Self-efficacy: The exercise of control*. Freedom and Company.

Banks, T., & Dohy, J. (2019). Mitigating barriers to persistence: A review of efforts to improve retention and graduation rates for students of color in higher education. *Higher Education Studies*, *9*(1), 118–131. https://doi.org/10.5539/hes.v9n1p118

Bempechat, J., & Shernoff, D. J. (2012). Parental influences on achievement motivation and student engagement. In S. Christenson, A. Reschly, & C. Wylie (Eds.), *Handbook of research on student engagement* (pp. 315–385). Springer. https://doi.org/10.1007/978-1-4614-2018-7_15

Bhutoria, A. (2022). Personalized education and artificial intelligence in the United States, China, and India: A systematic review using a human-in-the-loop model. *Computers and Education: Artificial Intelligence*, *3*, 100068. https://doi.org/10.1016/j.caeai.2022.100068

Bond, M., Bedenlier, S., Buntins, K., Kerres, M., & Zawacki-Richter, O. (2020). Facilitating student engagement in higher education through educational technology: A narrative systematic review in the field of education. *Contemporary Issues in Technology and Teacher Education*, *20*(2), 315–368.

Borger, J. G., Ng, A. P., Anderton, H., Ashdown, G. W., Auld, M., Blewitt, M. E., Brown, D. V., Call, M. J., Collins, P., & Freytag, S. (2023). Artificial intelligence takes center stage: Exploring the capabilities and implications of ChatGPT and other AI-assisted technologies in scientific research and education. *Immunology and Cell Biology*, *101*(10), 923–935. https://doi.org/10.1111/imcb.12689

Calais, J., & Green, M. (2022). The racial pandemic: Positive behavior intervention support as an asymptomatic carrier of racism. *Perspectives on Early Childhood Psychology and Education*, *6*(1), 4. https://doi.org/10.58948/2834-8257.1004

Calma, A., Cotronei-Baird, V., & Chia, A. (2022). Grammarly: An instructional intervention for writing enhancement in management education. *The International Journal of Management Education*, *20*(3), 100704. https://doi.org/10.1016/j.ijme.2022.100704

Camacho-Morles, J., Slemp, G. R., Pekrun, R., Loderer, K., Hou, H., & Oades, L. G. (2021). Activity achievement emotions and academic performance: A meta-analysis. *Educational Psychology Review*, *33*(3), 1051–1095. https://doi.org/10.1007/s10648-020-09585-3

Cassaretto, M., Espinosa, A., & Chau, C. (2024). Effects of resilience, social support, and academic self-efficacy, on mental health among Peruvian university students during the pandemic: The mediating role of digital inclusion. *Frontiers in Psychology*, *15*, 1282281. https://doi.org/10.3389/fpsyg.2024.1282281

Chen, X., Zou, D., Xie, H., Cheng, G., & Liu, C. (2022). Two decades of artificial intelligence in education. *Educational Technology & Society*, *25*(1), 28–47.

Cheng, S. L., Chang, J. C., Quilantan-Garza, K., & Gutierrez, M. L. (2023). Conscientiousness, prior experience, achievement emotions and academic procrastination in online learning environments. *British Journal of Educational Technology*, *54*(4), 898–923. https://doi.org/10.1111/bjet.13302

Clark, K. N., Dorio, N. B., Eldridge, M. A., Malecki, C. K., & Demaray, M. K. (2020). Adolescent academic achievement: A model of social support and grit. *Psychology in the Schools*, *57*(2), 204–221. https://doi.org/10.1002/pits.22318

Cotos, E. (2011). Potential of automated writing evaluation feedback. *Calico Journal*, *28*(2), 420–459. https://doi.org/10.11139/cj.28.2.420-459

Cotton, D. R., Cotton, P. A., & Shipway, J. R. (2023). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education and Teaching International*, 1–12.

Dahri, N. A., Yahaya, N., Al-Rahmi, W. M., Vighio, M. S., Alblehai, F., Soomro, R. B., & Shutaleva, A. (2024). Investigating AI-based academic support acceptance and its impact on students' performance in Malaysian and Pakistani higher education institutions. *Education and Information Technologies*, 1–50.

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., & Ahuja, M. (2023). Opinion paper: "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*, *71*, 102642. https://doi.org/10.1016/j.ijinfomgt.2023.102642

Elola, I., & Oskoz, A. (2017). Writing with 21st century social tools in the L2 classroom: New literacies, genres, and writing practices. *Journal of Second Language Writing*, *36*, 52–60. https://doi.org/10.1016/j.jslw.2017.04.002

Fitria, T. N. (2021). Quillbot as an online tool: Students' alternative in paraphrasing and rewriting of English writing. *Englisia: Journal of Language, Education, and Humanities*, *9*(1), 183–196. https://doi.org/10.22373/ej.v9i1.10233

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. *Review of Educational Research*, *74*(1), 59–109. https://doi.org/10.3102/00346543074001059

Getenet, S., Haeusler, C., Redmond, P., Cantle, R., & Crouch, V. (2024). First-year preservice teachers' understanding of digital technologies and their digital literacy, efficacy, attitude, and online learning engagement: Implication for course design. *Technology, Knowledge and Learning*, 1–25.

Goode, J. (2010). Mind the gap: The digital dimension of college access. *The Journal of Higher Education*, *81*(*5*), 583–618. https://doi.org/10.1080/00221546.2010.11779068

Goodfellow, R., & Lea, M. R. (2013). *Literacy in the digital university: Critical perspectives on learning, scholarship and technology*. Routledge.

Greenhow, C., Graham, C. R., & Koehler, M. J. (2022). Foundations of online learning: Challenges and opportunities. *Educational Psychologist*, *57*(3), 131–147. https://doi.org/10.1080/00461520.2022.2090364

Händel, M., Stephan, M., Gläser-Zikuda, M., Kopp, B., Bedenlier, S., & Ziegler, A. (2020). Digital readiness and its effects on higher education students' socio-emotional perceptions in the context of the COVID-19 pandemic. *Journal of Research on Technology in Education*, *54*(2), 267–280. https://doi.org/10.1080/15391523.2020.1846147

Ho, R. T., Lai, A. H., Lo, P. H., Nan, J. K., & Pon, A. K. (2017). A strength-based arts and play support program for young survivors in post-quake China: Effects on self-efficacy, peer support, and anxiety. *The Journal of Early Adolescence*, *37*(6), 805–824. https://doi.org/10.1177/0272431615624563

Hong, A. J., & Kim, H. J. (2018). College students' digital readiness for academic engagement (DRAE) scale: Scale development and validation. *The Asia-Pacific Education Researcher*, *27*(4), 303–312. https://doi.org/10.1007/s40299-018-0387-0

Hutson, J. (2024). Rethinking plagiarism in the era of generative AI. *Journal of Intelligent Communication*, *4*(1), 20–31-20–31. https://doi.org/10.54963/jic.v4i1.220

Iglesias-Pradas, S., Hernández-García, Á, Chaparro-Peláez, J., & Prieto, J. L. (2021). Emergency remote teaching and students' academic performance in higher education during the COVID-19 pandemic: A case study. *Computers in Human Behavior*, *119*, 106713. https://doi.org/10.1016/j.chb.2021.106713

Katsarou, E. (2021). The effects of computer anxiety and self-efficacy on L2 learners' self-perceived digital competence and satisfaction in higher education. *Journal of Education and E-Learning Research*, *8*(2), 158–172. https://doi.org/10.20448/journal.509.2021.82.158.172

Kohnke, L., & Moorhouse, B. L. (2021). Adopting HyFlex in higher education in response to COVID-19: Students' perspectives. *Open Learning: The Journal of Open, Distance and e-Learning*, *36*(3), 231–244. https://doi.org/10.1080/02680513.2021.1906641

Lee, Y.-F., Hwang, G.-J., & Chen, P.-Y. (2022). Impacts of an AI-based cha bot on college students' after-class review, academic performance, self-efficacy, learning attitude, and motivation. *Educational Technology Research and Development*, *70*(5), 1843–1865. https://doi.org/10.1007/s11423-022-10142-8

Lingard, L. (2023). Writing with ChatGPT: An illustration of its capacity, limitations & implications for academic writers. *Perspectives on medical education*, *12*(1), 261. https://doi.org/10.5334/pme.1072

Luan, L., Hong, J.-C., Cao, M., Dong, Y., & Hou, X. (2023). Exploring the role of online EFL learners' perceived social support in their learning engagement: A structural equation model. *Interactive Learning Environments*, *31*(3), 1703–1714. https://doi.org/10.1080/10494820.2020.1855211

Lund, B. D., Wang, T., Mannuru, N. R., Nie, B., Shimray, S., & Wang, Z. (2023). ChatGPT and a new academic reality: Artificial intelligence-written research papers and the ethics of the large language models in scholarly publishing. *Journal of the Association for Information Science and Technology*, *74*(5), 570–581. https://doi.org/10.1002/asi.24750

Malik, A. R., Pratiwi, Y., Andajani, K., Numertayasa, I. W., Suharti, S., & Darwis, A. (2023). Exploring artificial intelligence in academic essay: Higher education student's perspective. *International Journal of Educational Research Open*, *5*, 100296. https://doi.org/10.1016/j.ijedro.2023.100296

Martins, J., Cunha, J., Lopes, S., Moreira, T., & Rosário, P. (2022). School engagement in elementary school: A systematic review of 35 years of research. *Educational Psychology Review*, *34*(2), 793–849. https://doi.org/10.1007/s10648-021-09642-5

Marzuki, W., Rusdin, U., Darwin, D., & Indrawati, I. (2023). The impact of AI writing tools on the content and organization of students' writing: EFL teachers' perspective. *Cogent Education*, *10*(2), 2236469. https://doi.org/10.1080/2331186X.2023.2236469

Mitchell, K. M., McMillan, D. E., Lobchuk, M. M., Nickel, N. C., Rabbani, R., & Li, J. (2021). Development and validation of the situated academic writing self-efficacy scale (SAWSES). *Assessing Writing*, *48*, 100524. https://doi.org/10.1016/j.asw.2021.100524

Morley, J., Machado, C. C., Burr, C., Cowls, J., Joshi, I., Taddeo, M., & Floridi, L. (2020). The ethics of AI in health care: A mapping review. *Social Science & Medicine*, *260*, 113172. https://doi.org/10.1016/j.socscimed.2020.113172

Munoko, I., Brown-Liburd, H. L., & Vasarhelyi, M. (2020). The ethical implications of using artificial intelligence in auditing. *Journal of Business Ethics*, *167*(2), 209–234. https://doi.org/10.1007/s10551-019-04407-1

Nguyen, T. D., Cannata, M., & Miller, J. (2018). Understanding student behavioral engagement: Importance of student interaction with peers and teachers. *The Journal of Educational Research*, *111*(2), 163–174. https://doi.org/10.1080/00220671.2016.1220359

Nurseha, I. (2023). EFL teachers' perspectives on how AI writing tools affect the structure and substance of students' writing. *JEET, Journal of English Education and Technology*, *4*(03), 321–349.

Oflaz, A. (2019). The effects of anxiety, shyness and language learning strategies on speaking skills and academic achievement. *European Journal of Educational Research*, *8*(4), 999–1011. https://doi.org/10.12973/eu-jer.8.4.999

Parra, G. L., & Calero, S. X. (2019). Automated writing evaluation tools in the improvement of the writing skill. *International Journal of Instruction*, *12*(2), 209–226. https://doi.org/10.29333/iji.2019.12214a

Paul, N., Lin, T.-J., Ha, S. Y., Chen, J., & Newell, G. E. (2021). The role of achievement goal orientations in the relationships between high school students' anxiety, self-efficacy, and perceived use of revision strategies in argumentative writing. *Journal of Writing Research*, *12*(3), 657–684. https://doi.org/10.17239/jowr-2021.12.03.05

Pekrun, R., Goetz, T., Titz, W., & Perry, R. P. (2002). Academic emotions in students' self-regulated learning and achievement: A program of qualitative and quantitative research. *Educational Psychologist*, *37*(2), 91–105. https://doi.org/10.1207/S15326985EP3702_4

Pekrun, R., & Linnenbrink-Garcia, L. (2012). Academic emotions and student engagement. In S. Christenson, A. Reschly, & C. Wylie (Eds.), *Handbook of research on student engagement* (pp. 259–282). Springer. https://doi.org/10.1007/978-1-4614-2018-7_12

Pekrun, R., Marsh, H. W., Elliot, A. J., Stockinger, K., Perry, R. P., Vogl, E., Goetz, T., Van Tilburg, W. A., Lüdtke, O., & Vispoel, W. P. (2023). A three-dimensional taxonomy of achievement emotions. *Journal of Personality and Social Psychology*, *124*(1), 145. https://doi.org/10.1037/pspp0000448

Phyo, W. M., Nikolov, M., & Hódi, Á. (2023). Doctoral students' English academic writing experiences through metaphor analysis. *Heliyon*, *9*(2), e13293. https://doi.org/10.1016/j.heliyon.2023.e13293

Prunkl, C. E., Ashurst, C., Anderljung, M., Webb, H., Leike, J., & Dafoe, A. (2021). Institutionalizing ethics in AI through broader impact requirements. *Nature Machine Intelligence*, *3*(2), 104–110. https://doi.org/10.1038/s42256-021-00298-y

Qawqzeh, Y. (2024). Exploring the influence of student interaction with ChatGPT on critical thinking, problem solving, and creativity. *International Journal of Information and Education Technology*, *14*(4), 596–601. https://doi.org/10.18178/ijiet.2024.14.4.2082.

Rad, H. S., Alipour, R., & Jafarpour, A. (2024). Using artificial intelligence to foster students' writing feedback literacy, engagement, and outcome: A case of wordtune application. *Interactive Learning Environments*, *32*(9), 5020–5040. https://doi.org/10.1080/10494820.2023.2208170

Raheem, B. R., Anjum, F., & Ghafar, Z. N. (2023). Exploring the profound impact of artificial intelligence applications (Quillbot, Grammarly and ChatGPT) on English academic writing: A systematic review. *International Journal of Integrative Research (IJIR)*, *1*(10), 599–622.

Selvaraj, M., & Aziz, A. A. (2019). Systematic review: Approaches in teaching writing skill in ESL classrooms. *International Journal of Academic Research in Progressive Education and Development*, *8*(4), 450–473. https://doi.org/10.6007/IJARPED/v8-i4/6564

Song, C., & Song, Y. (2023). Enhancing academic writing skills and motivation: Assessing the efficacy of ChatGPT in AI-assisted language learning for EFL students. *Frontiers in Psychology*, *14*, 1260843. https://doi.org/10.3389/fpsyg.2023.1260843

Storey, V. A. (2023). AI technology and academic writing: Knowing and mastering the "Craft skills". *International Journal of Adult Education and Technology (IJAET)*, *14*(1), 1–15. https://doi.org/10.4018/IJAET.325795

Strobl, C., Ailhaud, E., Benetos, K., Devitt, A., Kruse, O., Proske, A., & Rapp, C. (2019). Digital support for academic writing: A review of technologies and pedagogies. *Computers & Education*, *131*, 33–48. https://doi.org/10.1016/j.compedu.2018.12.005

Taguma, M., Feron, E., & Lim, M. H. (2018). *Future of education and skills 2030: Conceptual learning framework*. Organization of Economic Co-operation and Development.

Tecedor, M. (2024). Digital storytelling: Changing learners' attitudes and self-efficacy beliefs. *Applied Linguistics*, *45*(1), 65–87. https://doi.org/10.1093/applin/amad002

Teng, L. S., Sun, P. P., & Xu, L. (2018). Conceptualizing writing self-efficacy in English as a foreign language contexts: Scale validation through structural equation modeling. *Tesol Quarterly*, *52*(4), 911–942. https://doi.org/10.1002/tesq.432

Usher, E. L., & Pajares, F. (2008). Sources of self-efficacy in school: Critical review of the literature and future directions. *Review of Educational Research*, *78*(4), 751–796. https://doi.org/10.3102/0034654308321456

Wang, C., Teng, M. F., & Liu, S. (2023). Psychosocial profiles of university students' emotional adjustment, perceived social support, self-efficacy belief, and foreign language anxiety during COVID-19. *Educational and Developmental Psychologist*, *40*(1), 51–62. https://doi.org/10.1080/20590776.2021.2012085

Zhao, D. (2024). The impact of AI-enhanced natural language processing tools on writing proficiency: An analysis of language precision, content summarization, and creative writing facilitation. *Education and Information Technologies*, 1–32.